The Distributed Cognition of a Fast Casual Restaurant

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Location of the observation: Qdoba in Boulder, CO

Date of the field study: September 22, 2024

Date of the assignment: September 26, 2024

Overview of project

- **Why did i choose this site?**

  I chose Qdoba in boulder, colorado as I've never been to a qdoba before. I've heard that they have various ingredients that we could add to the order. I thought this would be a good place to conduct this research in.

- **What did I expect to see here in terms of an information system? What assumptions did I have to try to leave at the door?**

  I had no clue about how different things were arranged, or what could be the different types of orders that could be placed. I expected it to be very similar to chipotle as they have the same theme.

Overview of project

- **What do they “produce?”**

  Qdoba is a chain of fast casual restaurant that produces mexican food. It could be dine in or takeaway. It mainly has bowls, burritos, and nachos.

- **What task did I observe?**

  I observed them taking orders and preparing them. People started their order in a line, they had to pick a type for a dish and then choose everything that would go in there. At the end, they had a counter for billing the orders. Here the product would be given to the customer, and they had to pay.

- **What was my role in this information system?**

  I was a customer getting food. I was also a person doing the research and collecting the data. I had talked to one of the employees before, and made sure it was okay to take pictures.

Method

- **The date and time of field study (list multiple visits), and pros and cons of this selection.**

  Sunday, September 22nd, at 4:50 pm

  Pros: going at this time gave me more time to observe the surroundings and take in more information about how things are placed. I went with one of my friends so that I could properly observe how things worked when someone else was ordering food.

  Cons: it would've been better if I had gone during lunch time, then I could've observed how they function with multiple and consecutive orders.

- **How did I observe?**

  I mostly took pictures and tried to take in as much information as I could. I wish I had a notepad with me to write stuff down when I was ordering. It was difficult to order the food, take pictures and get information at the same time.

- **What data did I collect?**

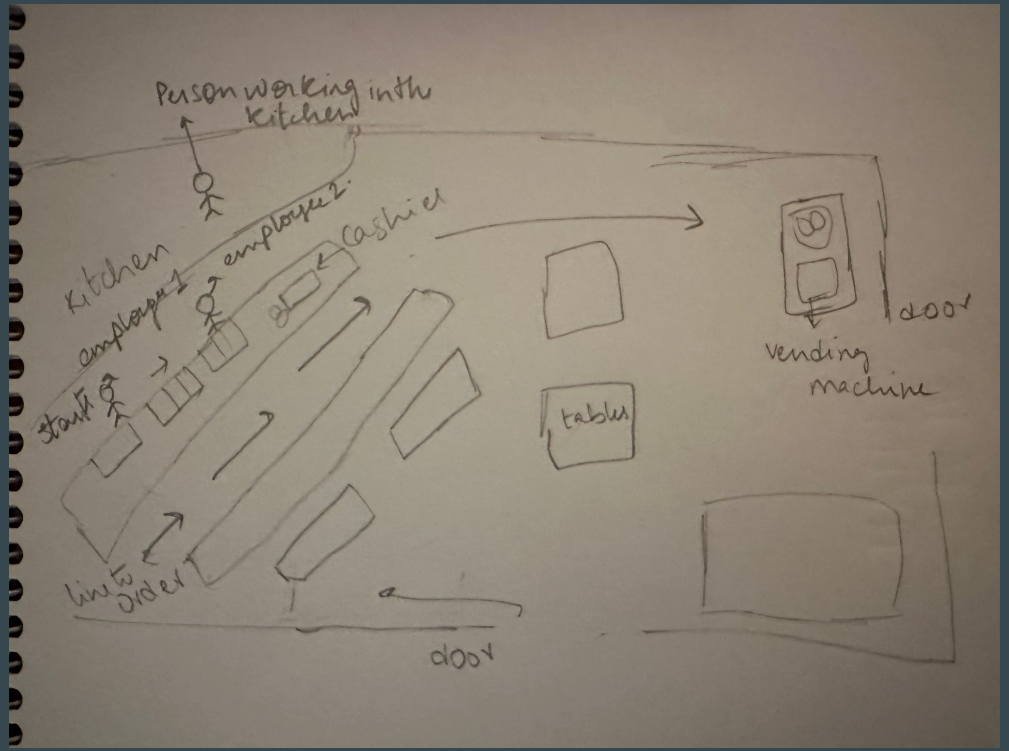
  I have collected pictures, and wrote down a bunch of points after I got my food. I also tried to draw a map of how the place is laid out.

RESEARCH DATA SECTION

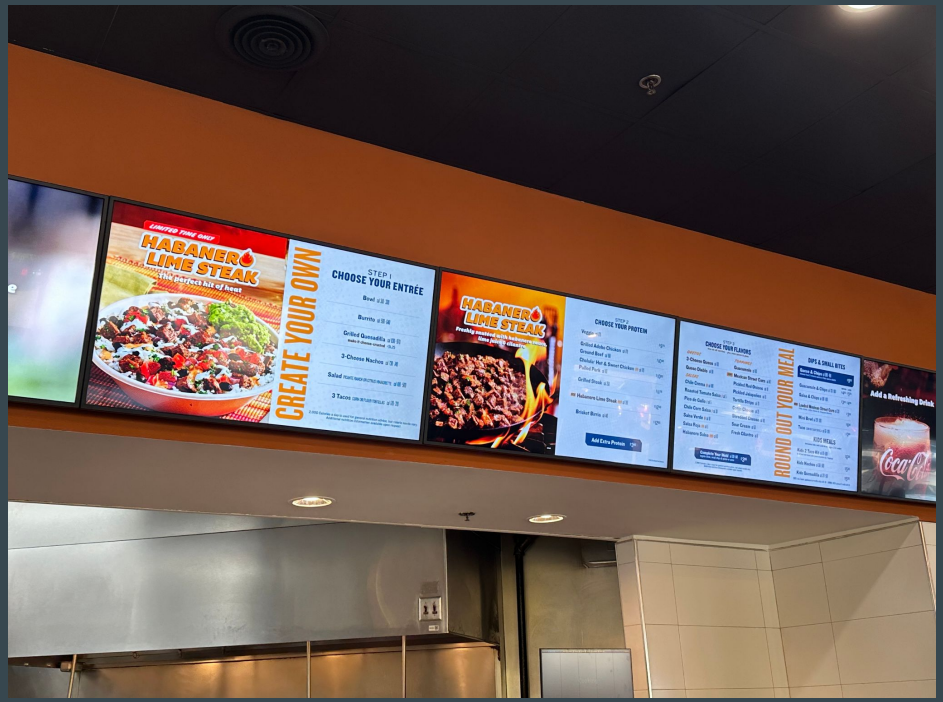
Description of information artifacts, division of labor, spatial arrangements, and other details that explain the task space.

Spatial layout





There is a line that the customers have to get into to start ordering their food. We can see the menu on the wall above the counter. The menu is organized into steps to make it easier for the people to think about what they would want to get. The first step is to choose your entree. The second step is to choose the protein you'd like to order. The third being the different flavors or add ons you could get.



The customer has to get into the line, and choose one of the entrees. A employee takes the required item for the food being made. In my case, a bowl. He then asked me the type of rice I'd like to pick for the base. Then he asked for the type of beans, meat and so on. All of these things were arranged in a specific order. They had this order laid out in front of them and the customers could see it too. The order of the items on the counter also matched the menu on the wall. As in, when I was standing at the place to pick the protein I could look up and see the menu and look for the type of items they had to offer. After the first employee completed the task to this level, he then just pushed the bowl to the next one. This one was already waiting for the food to get to him, and then continued to do the next steps. This employee asked me for the type of queso I'd like or if i'd want anything else from this area. Here we could see them laid out, but I didn't really know the names of the things present as they did not have them anywhere.





The next step was different types of salsas, and other toppings. There were wide variety of these present before us, but I was very confused with what to pick or how to ask the person for what I want. These last two steps felt very rushed to me. Then I realized that they actually had a little menu showing all of the available items, but it was very tiny and not as clearly visible to me. It was located on the bottom of the glass, and was tucked away in a corner.

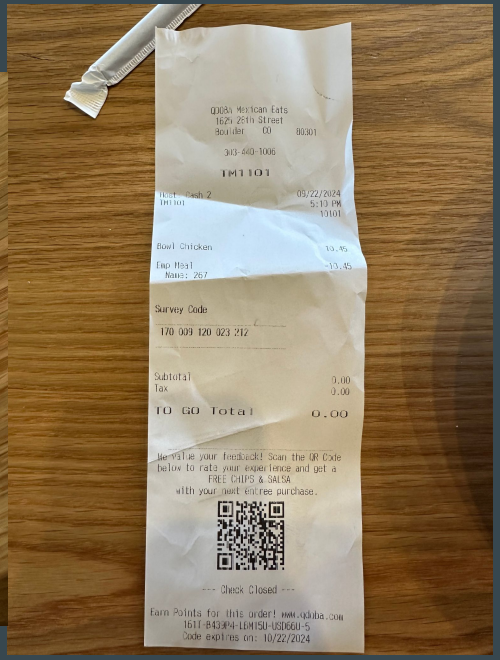






After the entire order was arranged, we got to the billing counter, the same employee that did the last two steps was doing this as well. I wonder if this would be different when there is a large line. As I could already see my friends order half done, and waiting for the next steps. As my friend's brother works at Qdoba he gave us both some coupons that we could use. This coupon said that we could have a free entree. It had his signature on it, and this was how the person billing the order recognized it. He still entered the items on to a bill and the just subtracted the amount by entering some code. I am not sure if they had different codes for coupons or a code that all the employees knew and had to enter in situations like these.





While at the counter we could order other food that they had to offer, for example, a kids meal or small bites. This was the place that we could also get drinks from. If we were to buy a drink, they would enter it into the bill. The person at the cashier saw the bowl and entered the necessary things onto the receipt. The drinks were charged according to the size, he gave us a cup, and we could walk to the vending machine and fill it up with any drink of our choice. This was the place where he hands the food and the receipt over to us.

As this was happening I could see the other employee taking other orders and this process happening all over again. They seemed to do all of that without communicating as much.



All of the ingredients were kept in a line, to make it easier for both the customer and the employee. All of the sections were clearly separated. They had a kitchen in the back where another employee was Preparing the food.

I could not see if they had any other arrangements that would make the process easier, or aid to the task.



The end product of the entire process



ANALYSIS

Social Space

- **Who are the actors involved?**

  The actors involved in this information system were customers, 3 employees. Two people that take orders, and a chef.

- **What is the division of labor?**
  - The customer is responsible for choosing their order and conveying it properly to the employee. They must follow the linear order in which the things work in the restaurant, and pay accordingly at the end.
  - The first employee is the one that is supposed to start off the order for the customer and go through the first steps with the them.
  - The second employee is the one that does the next steps of the process, and also takes care of the billing of the food.
  - The third employee is the one in charge of preparing the food.

Physical Space

- **How is the space arranged to make this happen?**

  The hand drawn map shows how the physical space was arranged. It was laid out in a way that everyone had space to do their task.

- **How are people arranged in it?**

  They were positioned in a way to help them know the order of the task being performed. They had their “designated” space to make the task easier and have a proper flow.

- **How are artifacts arranged in it (of all kinds—spatial distribution, charts, maps, reminders, labels)**

  All the ingredients were arranged in a linear way to make it easier for both the person plating and ordering food. The items were placed in the order that they were supposed to be chosen or put onto the bowl. I could not see any charts or maps that show any information. I am not sure if the employees used anything on their end, but it wasn't visible to the person ordering.

Temporal Space

- **Does time or timing place a role?**

  Since I was at the restaurant at the time when there weren't a lot of other people, I could not really see how the time played the role. But I do think the arrangement of the things made the task easier and took up a lot less time than it would if the things were arranged in a bad order. If there were more customers I think the employees would have gone through the tasks quicker to cater to the next people in the line.

Coordination

- **How does coordination happen for the task to be completed (between information propagations, between roles, across space, across time)?**

  I think the coordination between the employees is the key factor for the entire task. However, the spatial layout of different thing really helps with this too. Due to the placement of the menu and the ingredients corresponding to each other, it makes it easier for both the parties to understand each other and make sense of what step they are on.

  The switching of the roles between the employees was very interesting, they knew that everytime the first person passes the bowl to the next, he would have already completed the first steps and that this person has to continue from here.

  I would like to see how things would work if they were receiving orders online simultaneously.

Propagation of Representational States of Information

The person stands in the line and looks at the menu.

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The employee asks the person about what type of food they are going to get and starts plating it.

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They go through the first 2 steps together which is adding the rice, and protein. The customer can still see the menu on the wall to get a sense of what they have to offer.

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Then the first employee passes the food to the next one.

↓

This employee takes care of the next steps.

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At the end of counter, a cashier is located. They add all of the items chosen to the receipt. They also provide the customer with a cup if they ordered a drink

↓

The customer pays for the food and can ask for additional sauces if needed. They could also order other stuff on the menu. I did not have to pay because of the coupon. However, I had to hand it over to the employee as a form of payment.

↓

The drinks are available at the vending machine located in the restaurant.

Derivation of Design Recommendations or Principles

1. **Recommendation:** the labels for the salsas and other toppings were located at the bottom on the glass. I didn't even notice that they had them there until a bit later.

   It would have helped if they had little labels on the glass right where the items were located. It would make it easier for the people to both read the labels and also see the corresponding item.

2. **Principle:** I really liked how they had the menu on the wall. They items on the menu above corresponded to the things the person is supposed to pick in that step. Having this guide is really helpful for people that aren't regular customers.